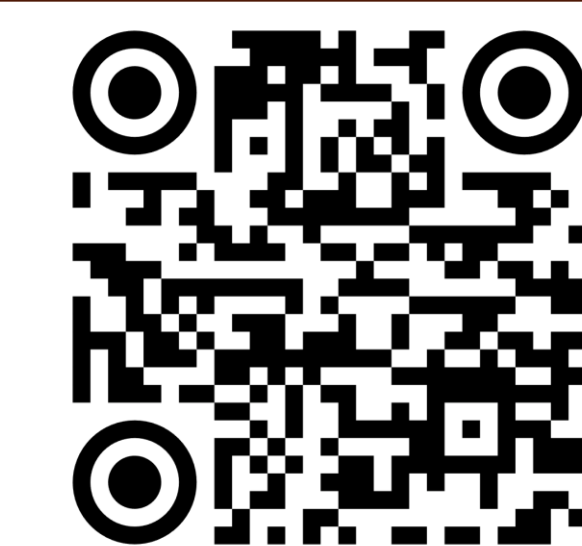
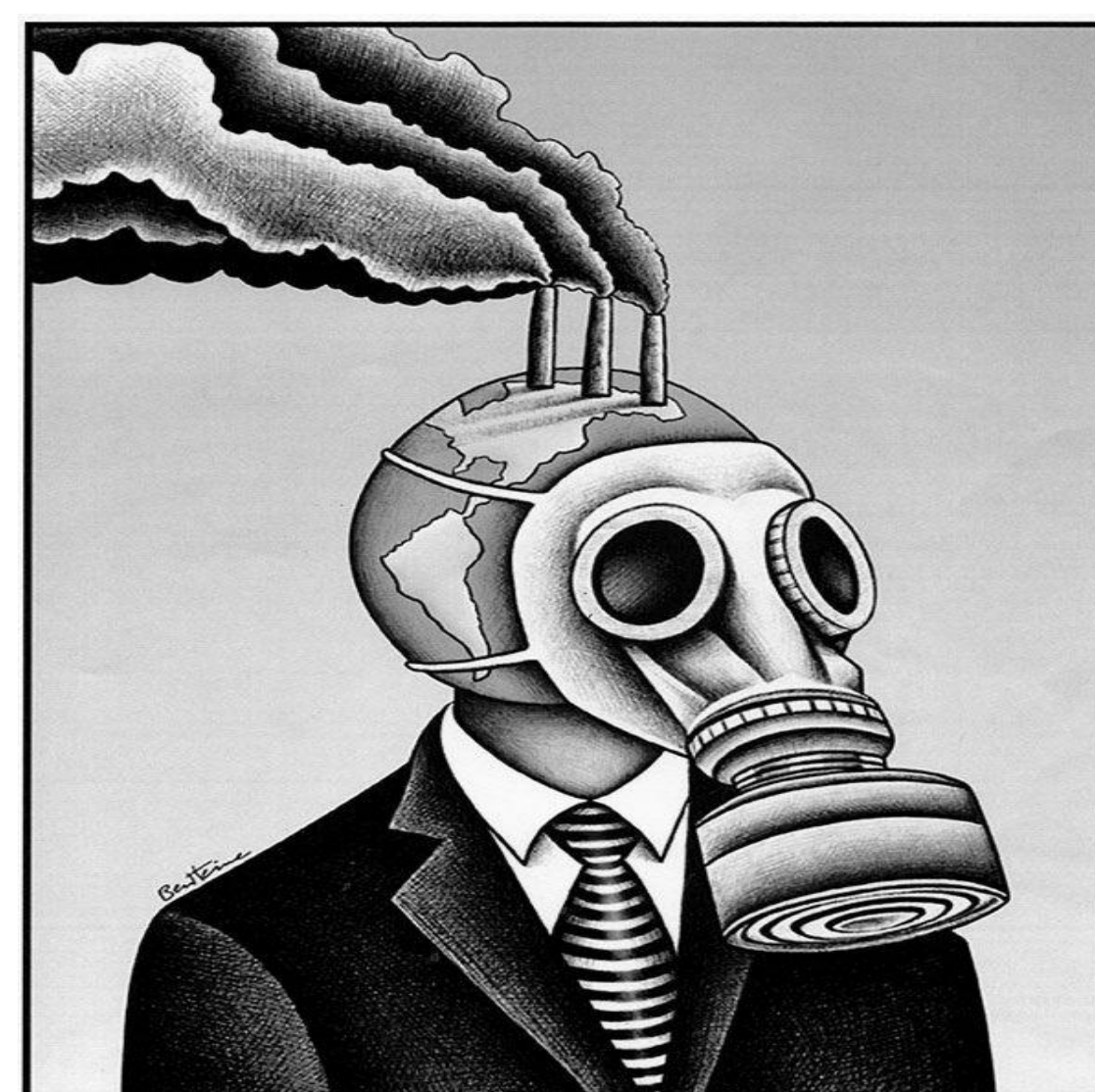


Global Air Quality Analysis: Exploring the Impact of Emissions, Forest Cover, and Population Density



Introduction

Air quality analysis explores the complex relationship between environmental and demographic factors across different nations, examining seven key metrics including air quality status, AQI values, greenhouse gas emissions, forest cover, and population density to understand global patterns of air pollution and identify potential correlations between environmental factors and air quality levels.



Research Question

How do such factors (Population density, GreenHouse Gasses Emissions, Forest percentage of lands) affect the AQI across countries and do mean AQI values differ significantly across continents?



Dataset

A global environmental snapshot with seven key metrics including air quality status, AQI values, greenhouse gas emissions, forest cover, and population density, showcasing the spectrum from high pollution levels to healthy air quality standards across different nations



Hypothesis

Hypothesis 1

H0: There is no correlation between AQI value and GHG emissions.
H1: There is a positive correlation between AQI value and GHG emissions.

Hypothesis 2

H0: There is no correlation between AQI value and forest cover.
H1: There is a negative correlation between AQI value and forest cover.

Hypothesis 3

H0: There is no correlation between AQI value and population density.
H1: There is a positive correlation between AQI value and population density.

Hypothesis 4

H0: There is no difference in mean AQI values among different continents.
H1: There is a significant difference in mean AQI values among different continents.



Conclusion

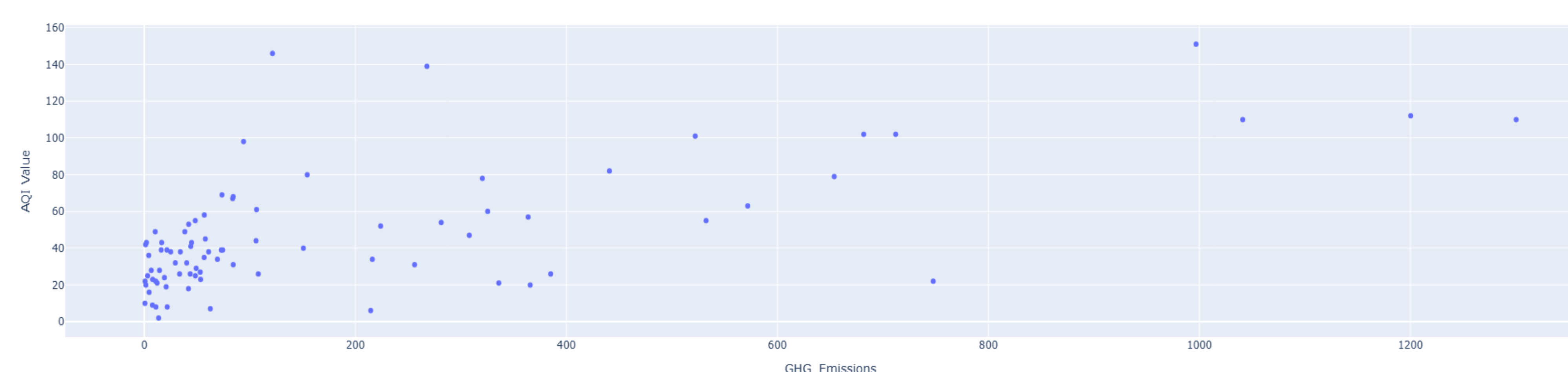
The comprehensive statistical analysis reveals a robust positive correlation between greenhouse gas emissions and Air Quality Index values ($r \approx 0.60$, $p < 0.05$), while demonstrating no significant relationships with forest cover or population density, and identifies substantial intercontinental variations in air quality metrics, particularly between Europe and Asia, indicating that industrial practices and regional environmental policies are the primary determinants of air quality outcomes. These findings underscore the critical importance of addressing industrial emissions as the principal strategy for improving air quality, while recognizing the necessity for region-specific approaches that account for local conditions, development patterns, and the complex interplay between environmental, economic, and technological factors.

Analysis

Analysis 1

There is a statistically significant positive correlation between GHG emissions and AQI values where higher GHG emissions are associated with higher AQI values (correlation coefficient of approximately 0.60, $p < 0.05$).

AQI Value vs. GHG Emissions (Correlation: 0.5060, p-value: 0.0000)



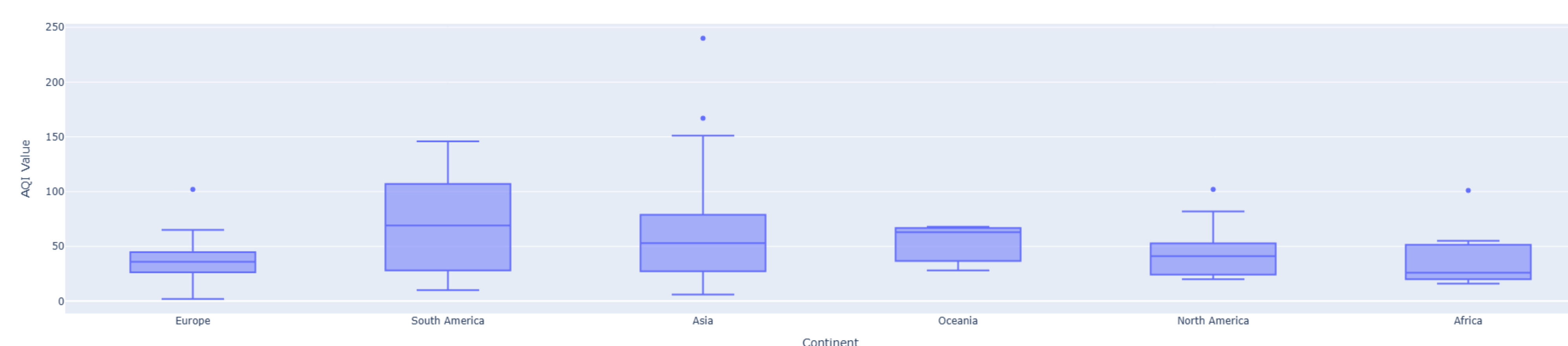
Analysis 2

There isn't a statistically significant correlation between forest cover and AQI values ($p > 0.05$)

Analysis 4

There is a statistically significant difference in AQI value means across different continents using the ANOVA test. Then we further used Tukey's HSD test to find which continents had the most significant difference and we found it to be between Europe and Asia ($p < 0.05$)

AQI Value Distribution Across Continents



Analysis 3

There isn't a statistically significant correlation between population density and AQI values ($p > 0.05$)



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